Examen 3

Date: *Wednesday, June 24, 2026*

1. (**5** ***points***) The reference frame K' rotates with respect to inertial (laboratory) reference frame K with angular velocity $\boldsymbol{\Omega}$. A pointlike particle has coordinates and momenta $(\boldsymbol{q}, \boldsymbol{p})$ in the frame K', and coordinates and momenta $(\boldsymbol{Q}, \boldsymbol{P})$ in the laboratory frame K.

   (a) (**2** ***points***) Write out explicitly the transformation which relates $(\boldsymbol{q}, \boldsymbol{p})$ and $(\boldsymbol{Q}, \boldsymbol{P})$, and demonstrate that it is canonical. Find explicitly the generating function $F_2(\boldsymbol{q},\, \boldsymbol{P})$ for this transformation.

   (b) (**1** ***point***) Assume that in the lab frame a particle moves in the field of the spherically symmetric harmonic oscillator $U = k\boldsymbol{Q}^2/2$. Find the hamiltonian $\tilde{H}(\boldsymbol{q}, \boldsymbol{p})$ of the system in the rotating frame.

   (c) (**2** ***points***) Write out the canonical equations of motion in terms of the variables $(\boldsymbol{q}, \boldsymbol{p})$ in the rotating frame and demonstrate they include contributions corresponding to the fictitious forces discussed in Lectures 10,11.

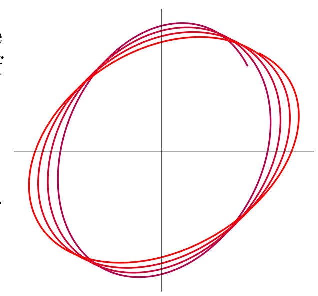
2. (***5 points***) A pointlike particle of mass m and charge q moves in the Coulomb field $U = -k/r$ and a homogeneous magnetic field $\boldsymbol{B} = \text{const}$. The interaction with magnetic field is very weak and may be considered as a small perturbation.

   - **(3** ***points*****)** Using canonical perturbation theory, analyze how the magnetic field affects the trajectory of the particle and find explicitly the first $\mathcal{O}(B)$-correction to it.

   - **(*2 points*)** It is known that in the limit $\Delta H = 0$ the trajectories in the Kepler problem are plain and closed, so the motion is periodic. Analyze if these properties remain valid in presence of perturbation $\Delta H \neq 0$.

     – If yes, find the correction to the period of motion of the particle.

     – If not, find the time-averaged corrections to the shape, size and orientation of the Keplerian orbit, induced by the perturbation ΔH.



3. (***5 points***) A nonrelativistic particle of mass m moves in one-dimensional symmetric power-law potential

$$U(x) = g|x|^{\alpha}, \quad g > 0, \quad \alpha > 0 \tag{1}$$

   (a) (***2 points***) Write out the corresponding canonical Hamiltonian and the Hamilton-Jacobi (HJ) equations. Resolve both equations and demonstrate that they give the same trajectories.

      i. (***1 point***) Assume that the coupling g changes very slowly (adiabatically). Analyze how the particle's energy $E(t)$ will change as a function of time.

      ii. (***1 point***) Determine how the period of oscillation $T(t)$ scales with $g(t)$ during this adiabatic transformation.

         A. (***1 point***) Show that your results in the limit $\alpha \to 2$ reproduce the well-known scaling laws for the harmonic oscillator.